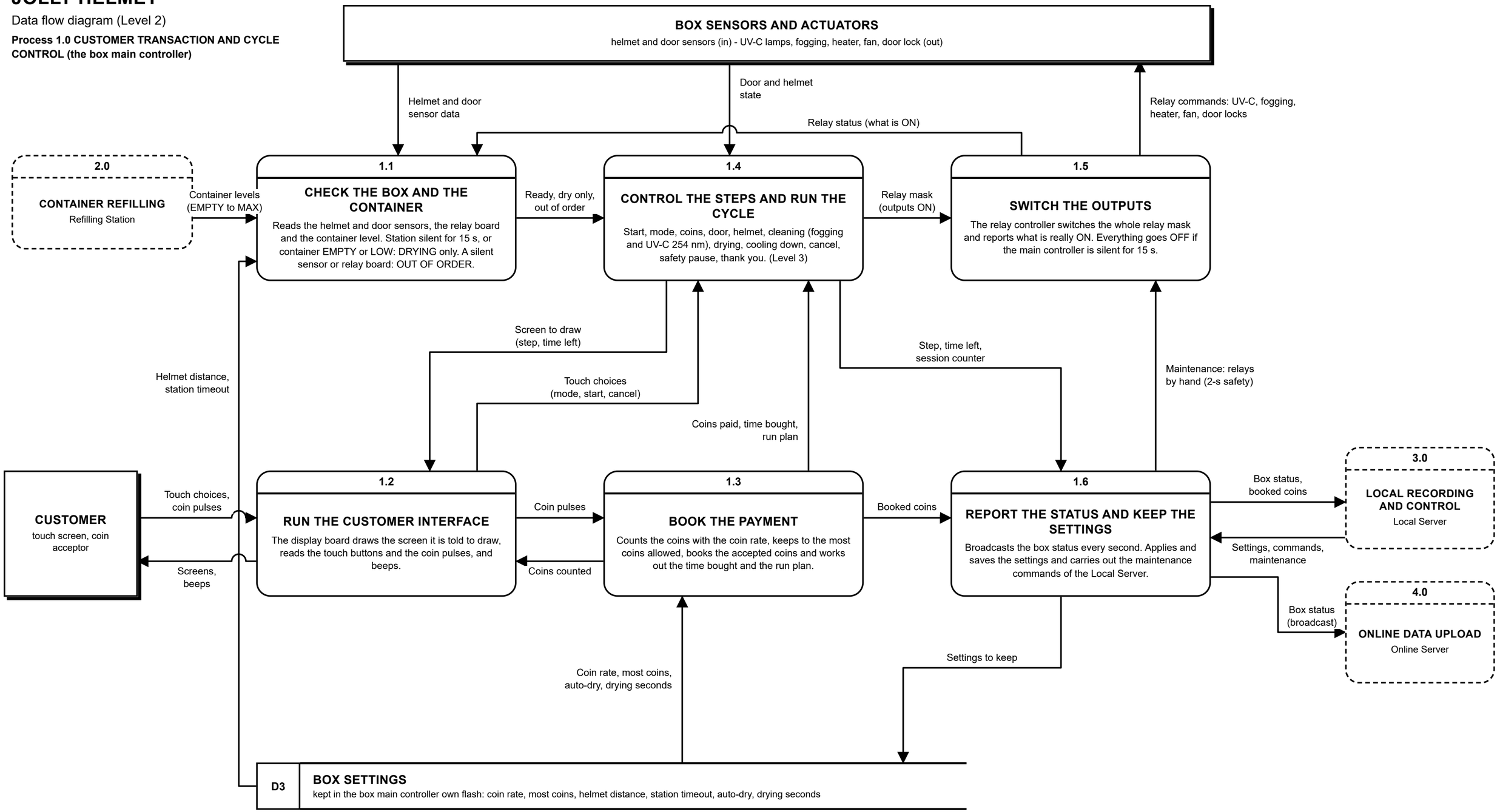
JOLLY HELMET

Data flow diagram (Level 2)

Process 1.0 CUSTOMER TRANSACTION AND CYCLE CONTROL (the box main controller)



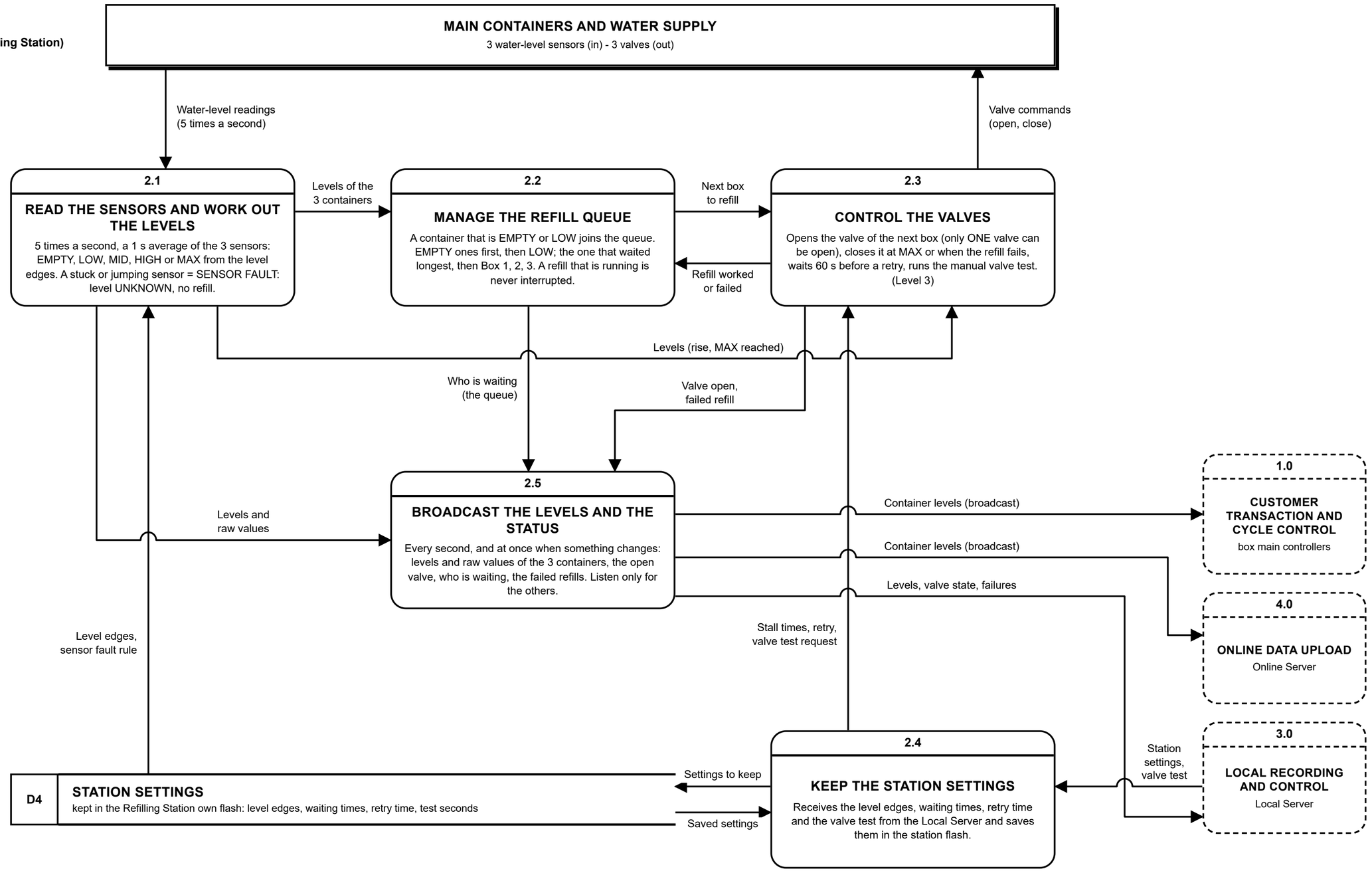
External entity · Process of this level · Other process (from Level 1 or 2) · Data store · ▶ Data flow

Level 2 splits one Level 1 process into its sub-processes. The data flows that enter and leave the dashed boxes are the same as in Level 1 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).

JOLLY HELMET

Data flow diagram (Level 2)

Process 2.0 CONTAINER REFILLING (the Refilling Station)



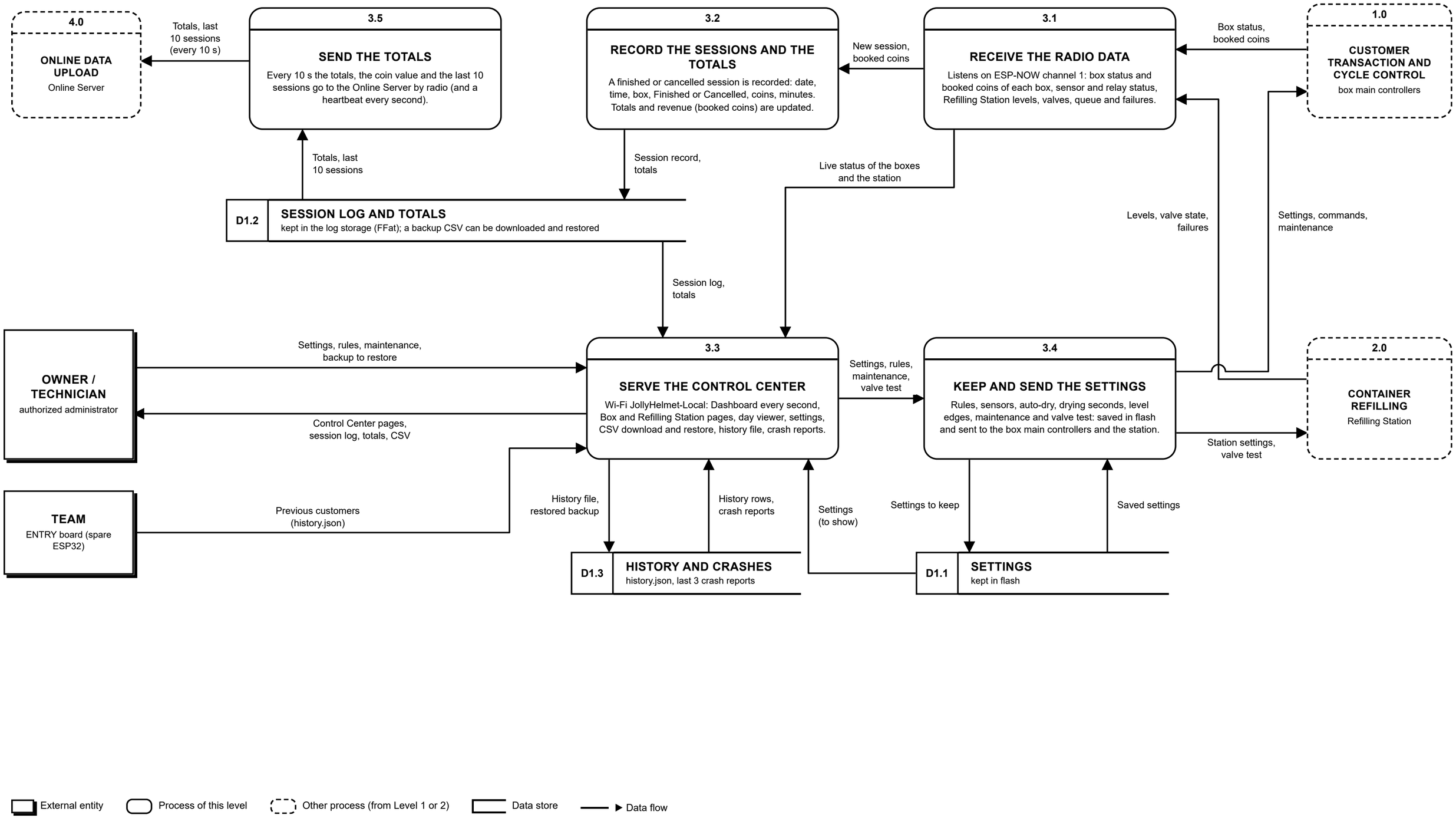
External entity · Process of this level · Other process (from Level 1 or 2) · Data store · ▶ Data flow

Level 2 splits one Level 1 process into its sub-processes. The data flows that enter and leave the dashed boxes are the same as in Level 1 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).

JOLLY HELMET

Data flow diagram (Level 2)

Process 3.0 LOCAL RECORDING AND CONTROL (the Local Server)

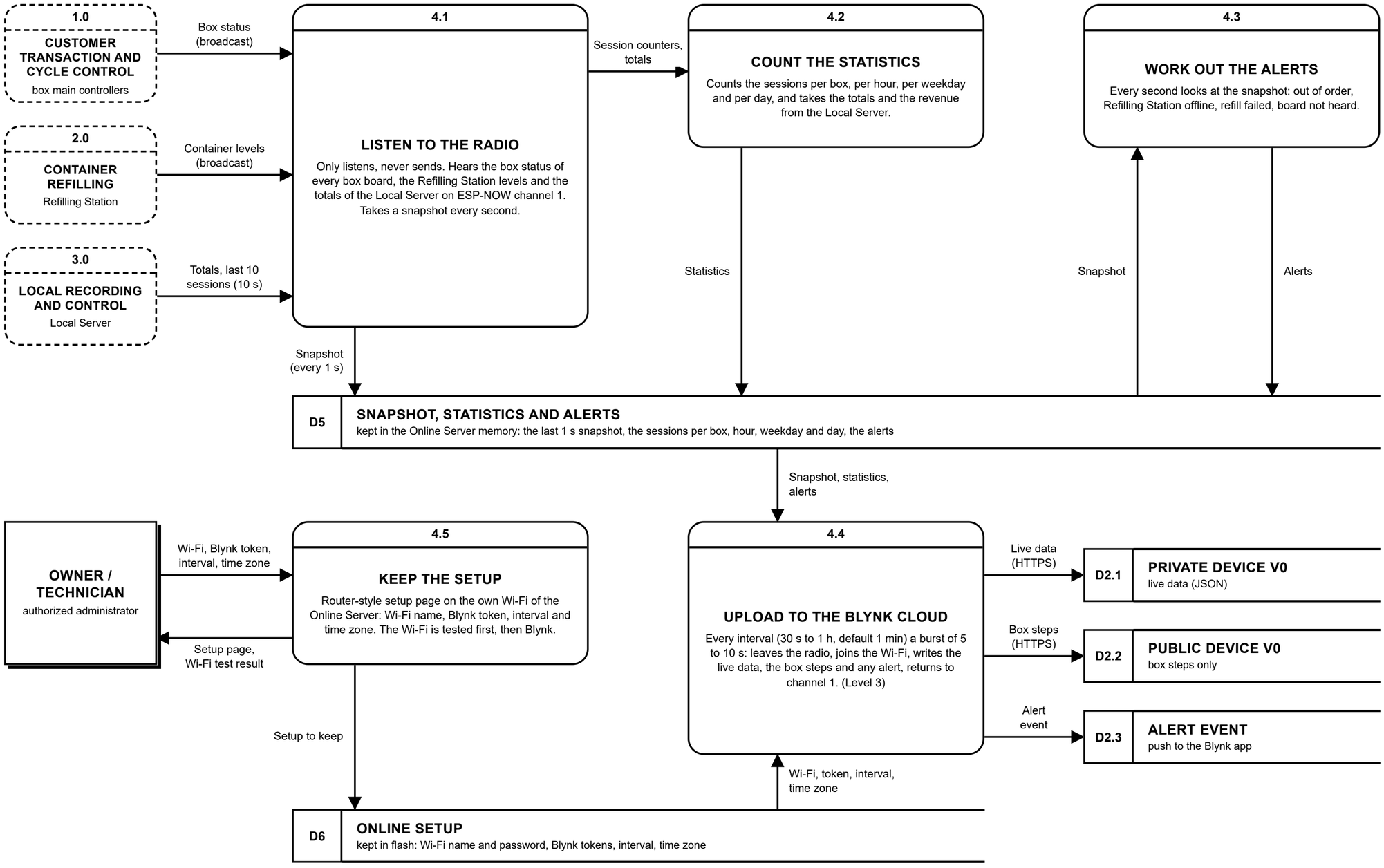


Level 2 splits one Level 1 process into its sub-processes. The data flows that enter and leave the dashed boxes are the same as in Level 1 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).

JOLLY HELMET

Data flow diagram (Level 2)

Process 4.0 ONLINE DATA UPLOAD (the Online Server)



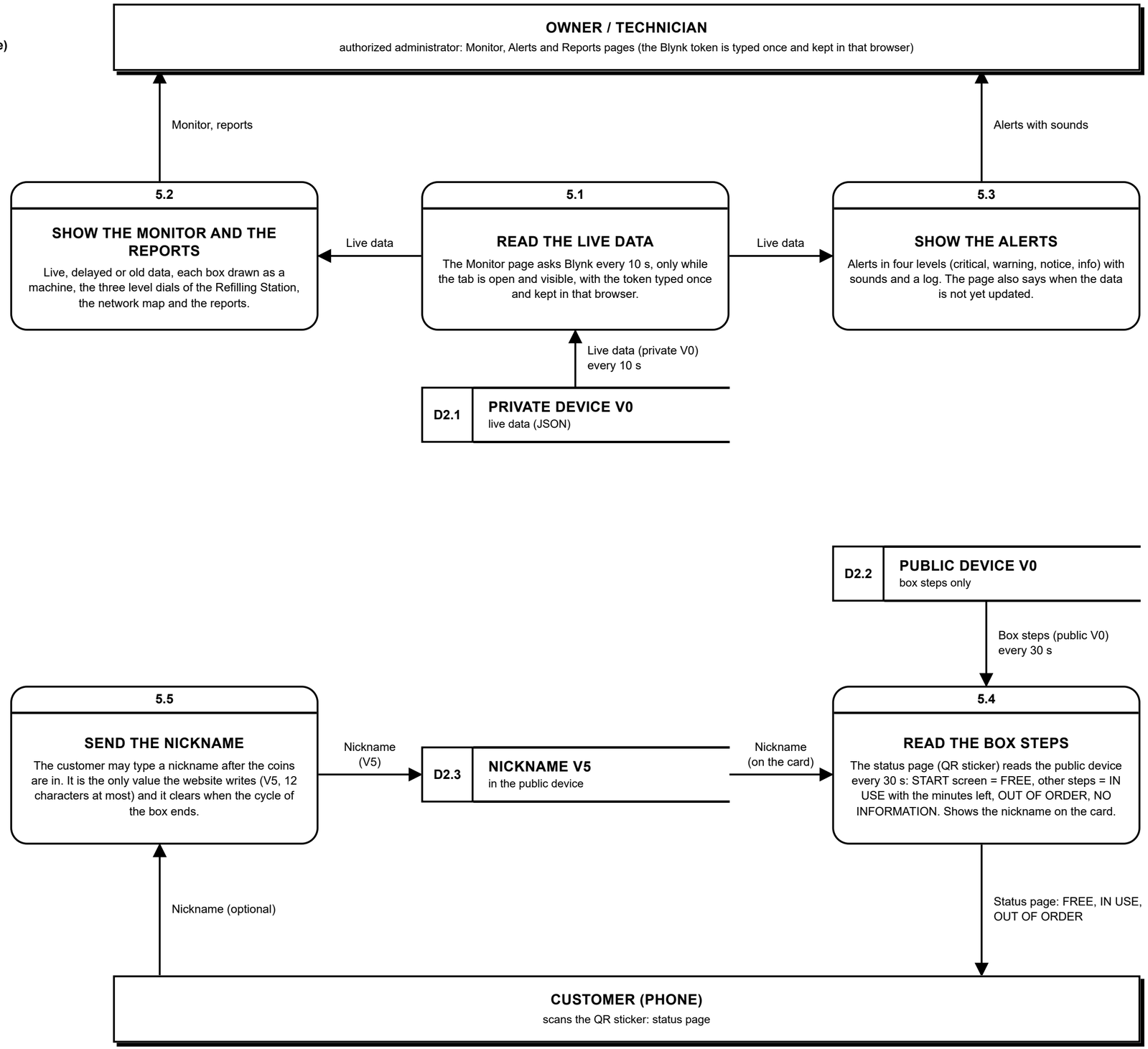
External entity · Process of this level · Other process (from Level 1 or 2) · Data store · ▶ Data flow

Level 2 splits one Level 1 process into its sub-processes. The data flows that enter and leave the dashed boxes are the same as in Level 1 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).

JOLLY HELMET

Data flow diagram (Level 2)

Process 5.0 ONLINE MONITORING (the website)



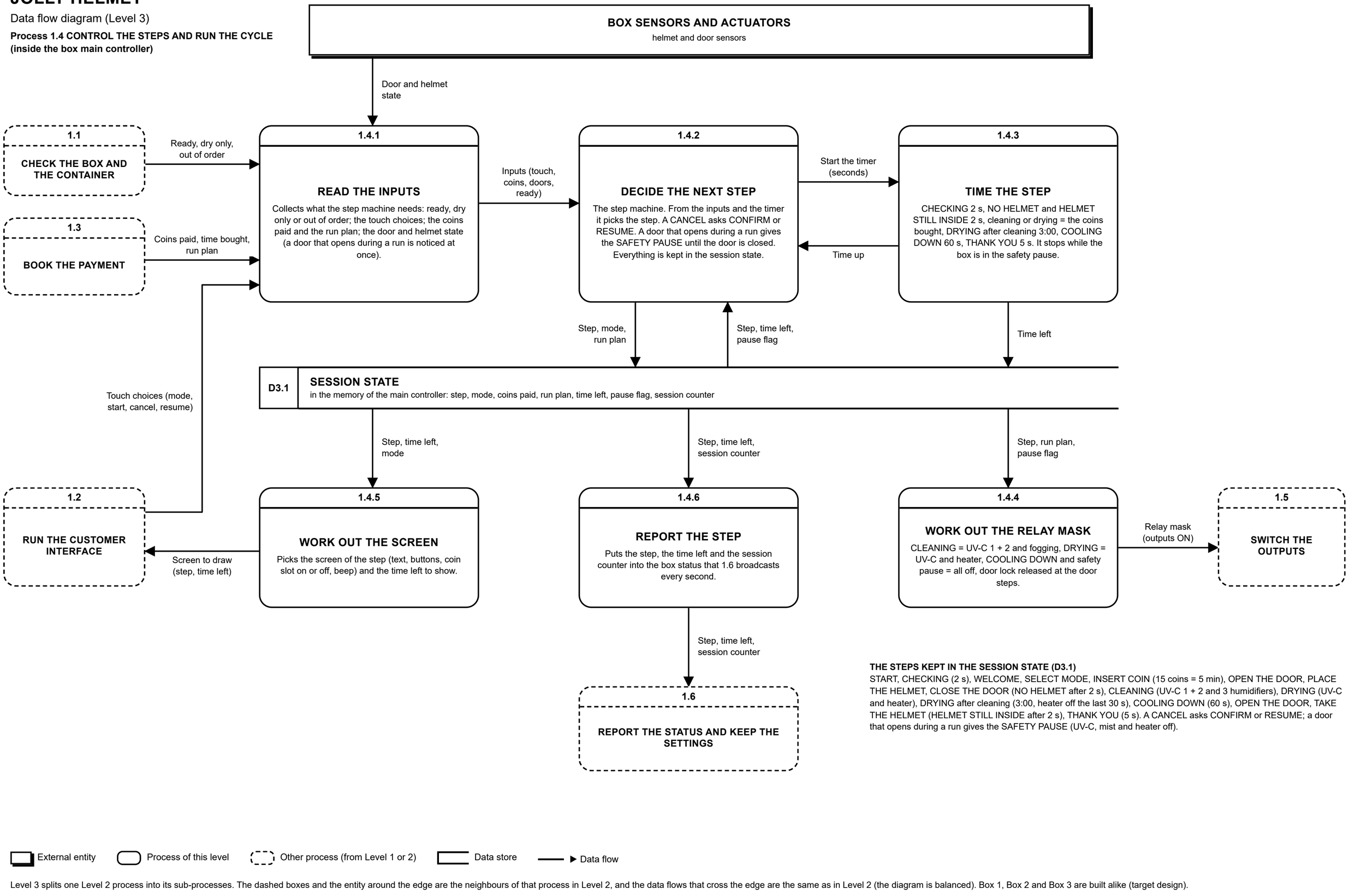
External entity · Process of this level · Other process (from Level 1 or 2) · Data store · ▶ Data flow

Level 2 splits one Level 1 process into its sub-processes. The data flows that enter and leave the dashed boxes are the same as in Level 1 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).

JOLLY HELMET

Data flow diagram (Level 3)

Process 1.4 CONTROL THE STEPS AND RUN THE CYCLE (inside the box main controller)



THE STEPS KEPT IN THE SESSION STATE (D3.1)
START, CHECKING (2 s), WELCOME, SELECT MODE, INSERT COIN (15 coins = 5 min), OPEN THE DOOR, PLACE THE HELMET, CLOSE THE DOOR (NO HELMET after 2 s), CLEANING (UV-C 1 + 2 and 3 humidifiers), DRYING (UV-C and heater), DRYING after cleaning (3:00, heater off the last 30 s), COOLING DOWN (60 s), OPEN THE DOOR, TAKE THE HELMET (HELMET STILL INSIDE after 2 s), THANK YOU (5 s). A CANCEL asks CONFIRM or RESUME; a door that opens during a run gives the SAFETY PAUSE (UV-C, mist and heater off).

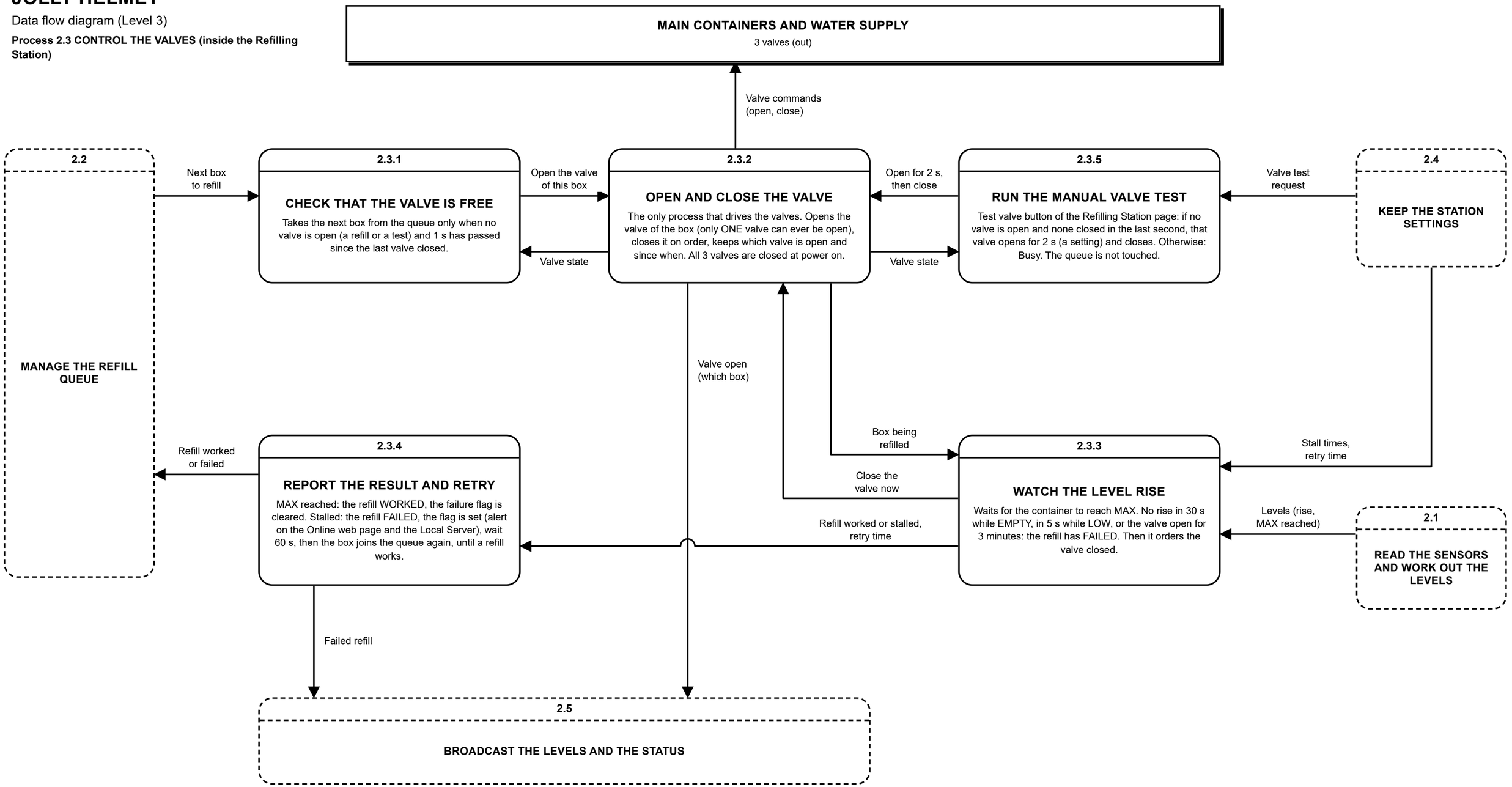
External entity · Process of this level · Other process (from Level 1 or 2) · Data store · ▶ Data flow

Level 3 splits one Level 2 process into its sub-processes. The dashed boxes and the entity around the edge are the neighbours of that process in Level 2, and the data flows that cross the edge are the same as in Level 2 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).

JOLLY HELMET

Data flow diagram (Level 3)

Process 2.3 CONTROL THE VALVES (inside the Refilling Station)



External entity · Process of this level · Other process (from Level 1 or 2) · Data store · ▶ Data flow

Level 3 splits one Level 2 process into its sub-processes. The dashed boxes and the entity around the edge are the neighbours of that process in Level 2, and the data flows that cross the edge are the same as in Level 2 (the diagram is balanced). Box 1, Box 2 and Box 3 are built alike (target design).